

ABS radio interface

## Functions of Abis Interface

1. Transfers voice and data signals
2. Carries control and signaling messages
3. Manages radio resources
4. Supports handover process
5. Connects multiple BTS to one BSC

## Features

- Usually a **wired connection** (optical fiber or microwave link)
- Uses **LAPD protocol** for signaling
- Supports communication between several BTS units

Differentiate between 2G and 3G

| Feature            | 2G                                                   | 3G                                                       |
|--------------------|------------------------------------------------------|----------------------------------------------------------|
| Full Form          | Second Generation                                    | Third Generation                                         |
| Technology Example | <b>Global System for Mobile Communications (GSM)</b> | <b>Universal Mobile Telecommunications System (UMTS)</b> |
| Data Speed         | Low (up to 64 kbps)                                  | High (up to 2 Mbps)                                      |
| Switching Type     | Circuit Switching                                    | Packet + Circuit Switching                               |
| Services           | Voice calls, SMS                                     | Voice, Video call, Internet                              |
| Internet           | Very slow                                            | Fast browsing                                            |
| Multimedia Support | Limited                                              | High (video streaming)                                   |
| Security           | Basic                                                | Improved security                                        |